

```
from google.colab import drive
drive.mount('/content/drive')
```

Mounted at /content/drive

```
import os
```

```
os.listdir('/content/drive/MyDrive')
```

```
['Classroom',
 "janani's certificates (2).pdf",
 "janani's certificates (1).pdf",
 'Poster sale.gsheet',
 'Detailed poster sale.gsheet',
 'one mark(DM) .pdf',
 'Polaroids sales.gsheet',
 'janani T.pptx',
 "janani's certificates.pdf",
 'college id.pdf',
 'janani photo.pdf',
 'fees receipt.pdf',
 'voter id (1).pdf',
 'voter id.pdf',
 'fees receipt 1.pdf',
 'pass book.pdf',
 'fees bonafide.pdf',
 'homepdf.pdf',
 'Team Insight 1.pptx',
 'Women empowerment.jpg',
 'My vote my rights.png',
 'Your paragraph text_20260317_183021_0000.png',
 'Python Programming unit-3,4.gdoc',
 'Colab Notebooks',
 'Elevate_labs-Task 1',
 'Netflix_titles.csv',
 'Screenshot_20261003_133352_GPay (1).jpg',
 'Screenshot_20261003_133352_GPay.jpg',
 'sales_data.csv',
 'Task2_Sales_Visual_Report.pdf',
 'task3_sales.db',
 'Titanic.csv']
```

```
import seaborn as sns
import pandas as pd
```

```
df = sns.load_dataset('titanic')
```

```
df.head()
```

	survived	pclass	sex	age	sibsp	parch	fare	embarked	class	who	adult_male
0	0	3	male	22.0	1	0	7.2500	S	Third	man	True
1	1	1	female	38.0	1	0	71.2833	C	First	woman	False
2	1	3	female	26.0	0	0	7.9250	S	Third	woman	False
3	1	1	female	35.0	1	0	53.1000	S	First	woman	False
4	0	3	male	35.0	0	0	8.0500	S	Third	man	True

Next steps: [Generate code with df](#) [New interactive sheet](#)

```
df.to_csv('/content/drive/MyDrive/Titanic.csv', index=False)
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 15 columns):
#   Column          Non-Null Count  Dtype
---  -
0   survived        891 non-null    int64
1   pclass          891 non-null    int64
2   sex             891 non-null    object
3   age            714 non-null    float64
4   sibsp          891 non-null    int64
5   parch          891 non-null    int64
6   fare           891 non-null    float64
7   embarked       889 non-null    object
8   class          891 non-null    category
9   who            891 non-null    object
10  adult_male     891 non-null    bool
11  deck          203 non-null    category
12  embark_town    889 non-null    object
13  alive          891 non-null    object
14  alone          891 non-null    bool
dtypes: bool(2), category(2), float64(2), int64(4), object(5)
memory usage: 80.7+ KB
```

```
df.describe()
```

	survived	pclass	age	sibsp	parch	fare	
count	891.000000	891.000000	714.000000	891.000000	891.000000	891.000000	
mean	0.383838	2.308642	29.699118	0.523008	0.381594	32.204208	
std	0.486592	0.836071	14.526497	1.102743	0.806057	49.693429	
min	0.000000	1.000000	0.420000	0.000000	0.000000	0.000000	
25%	0.000000	2.000000	20.125000	0.000000	0.000000	7.910400	
50%	0.000000	3.000000	28.000000	0.000000	0.000000	14.454200	
75%	1.000000	3.000000	38.000000	1.000000	0.000000	31.000000	
max	1.000000	3.000000	80.000000	8.000000	6.000000	512.329200	

```
df.isnull().sum()
```

	0
survived	0
pclass	0
sex	0
age	177
sibsp	0
parch	0
fare	0
embarked	2
class	0
who	0
adult_male	0
deck	688
embark_town	2
alive	0
alone	0

dtype: int64

```
df['survived'].value_counts()
```

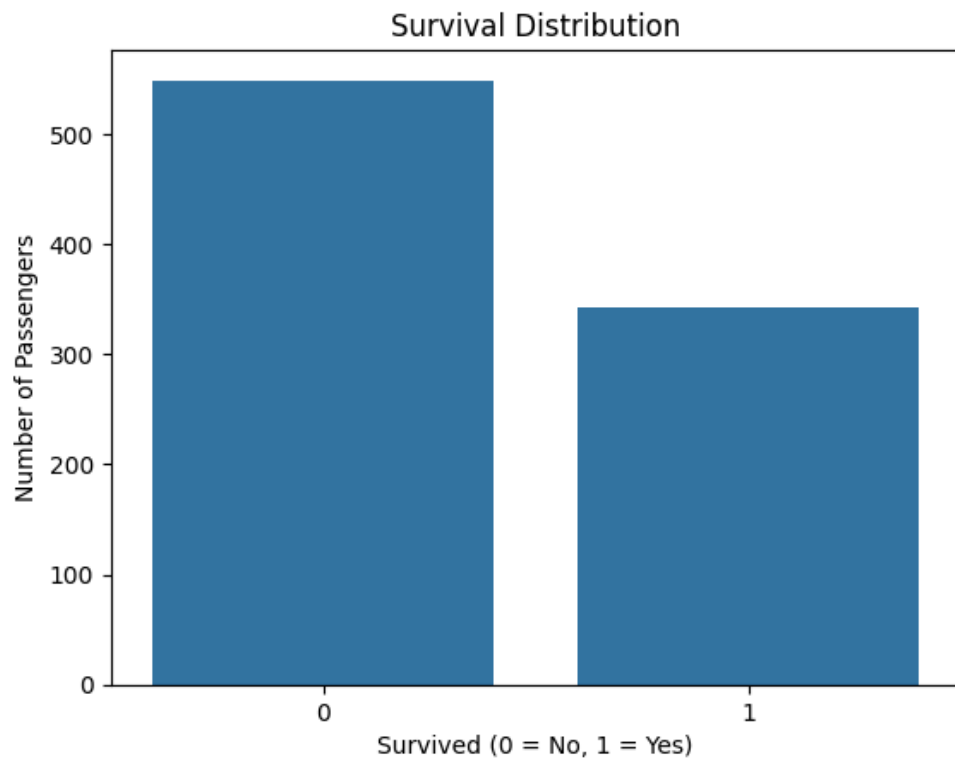
	count
survived	
0	549
1	342

dtype: int64

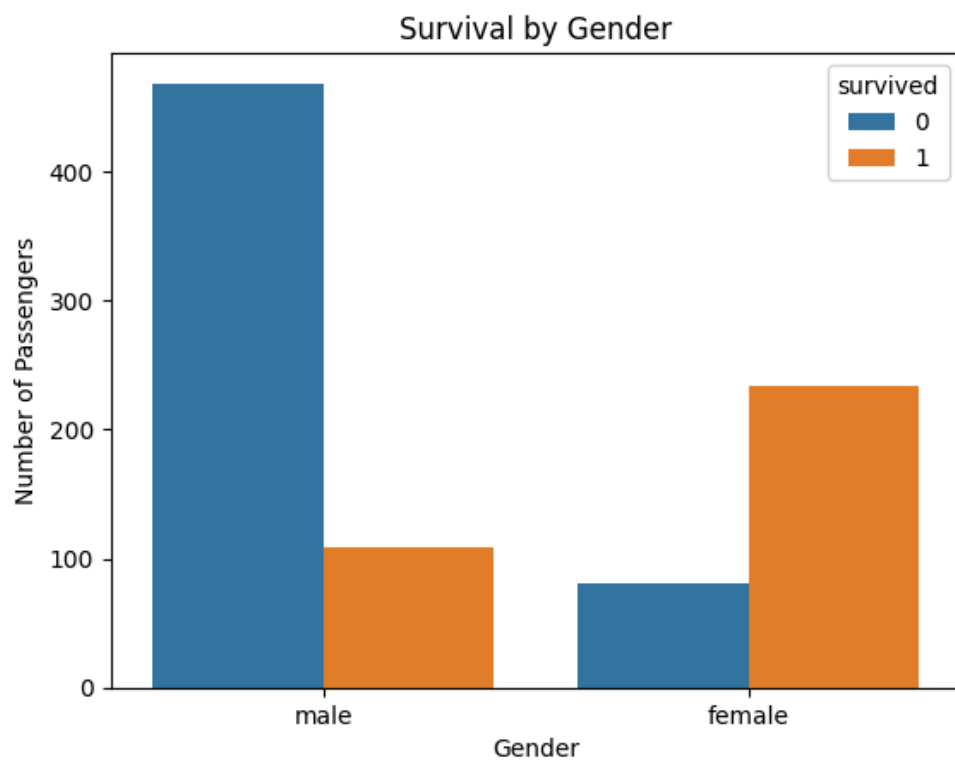
```
sns.countplot(x='survived', data=df)

plt.title('Survival Distribution')
plt.xlabel('Survived (0 = No, 1 = Yes)')
plt.ylabel('Number of Passengers')

plt.show()
```



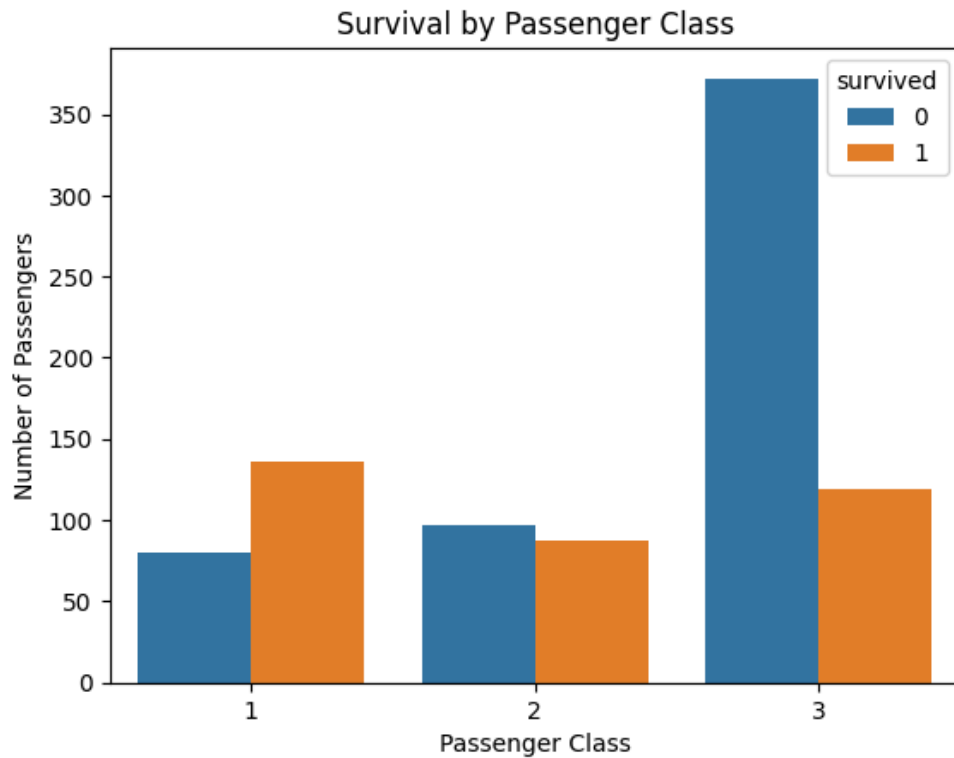
```
sns.countplot(x='sex', hue='survived', data=df)  
  
plt.title('Survival by Gender')  
plt.xlabel('Gender')  
plt.ylabel('Number of Passengers')  
  
plt.show()
```



```
sns.countplot(x='pclass', hue='survived', data=df)
```

```
plt.title('Survival by Passenger Class')  
plt.xlabel('Passenger Class')  
plt.ylabel('Number of Passengers')
```

```
plt.show()
```

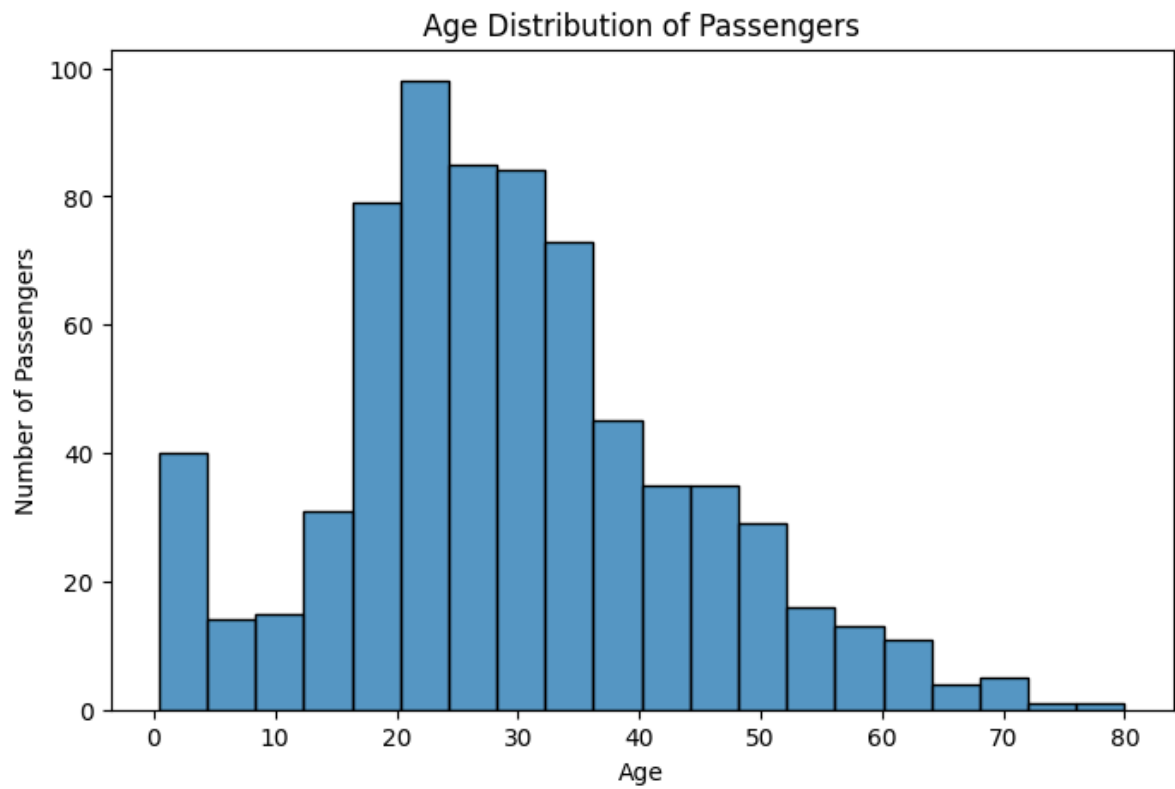


```
plt.figure(figsize=(8,5))
```

```
sns.histplot(df['age'].dropna(), bins=20)
```

```
plt.title('Age Distribution of Passengers')  
plt.xlabel('Age')  
plt.ylabel('Number of Passengers')
```

```
plt.show()
```



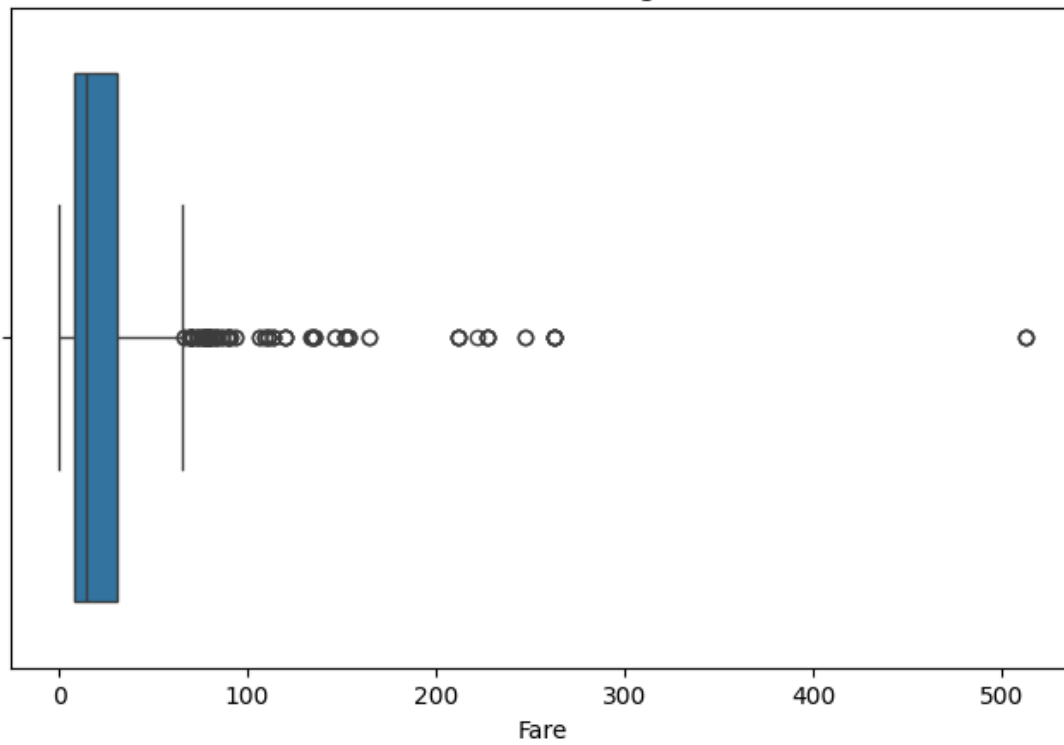
```
plt.figure(figsize=(8,5))

sns.boxplot(x=df['fare'])

plt.title('Distribution of Passenger Fare')
plt.xlabel('Fare')

plt.show()
```

Distribution of Passenger Fare

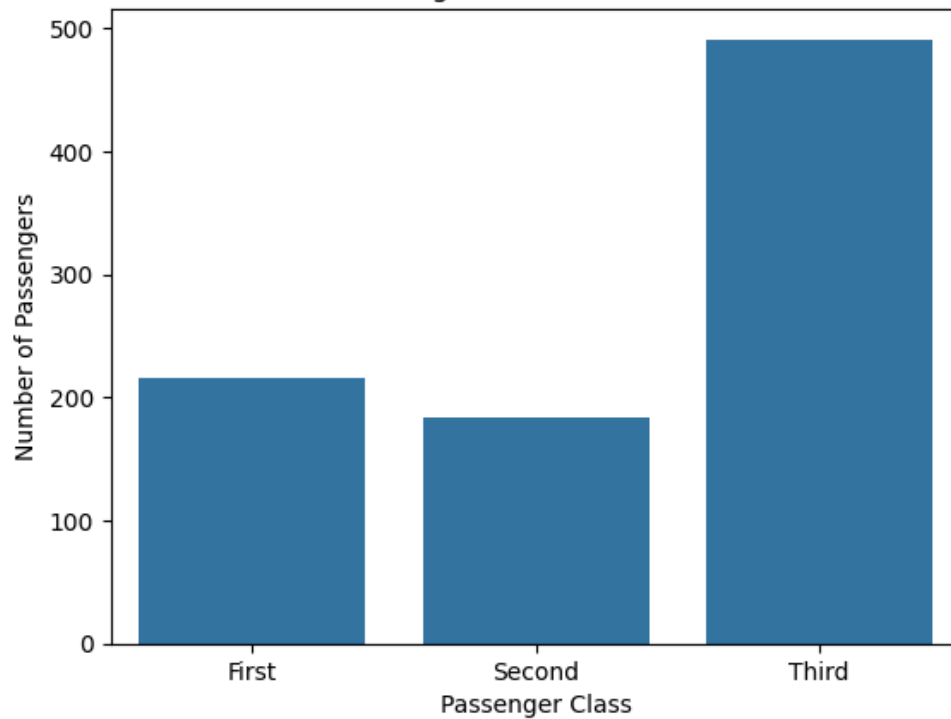


```
sns.countplot(x='class', data=df)

plt.title('Passenger Class Distribution')
plt.xlabel('Passenger Class')
plt.ylabel('Number of Passengers')

plt.show()
```

Passenger Class Distribution



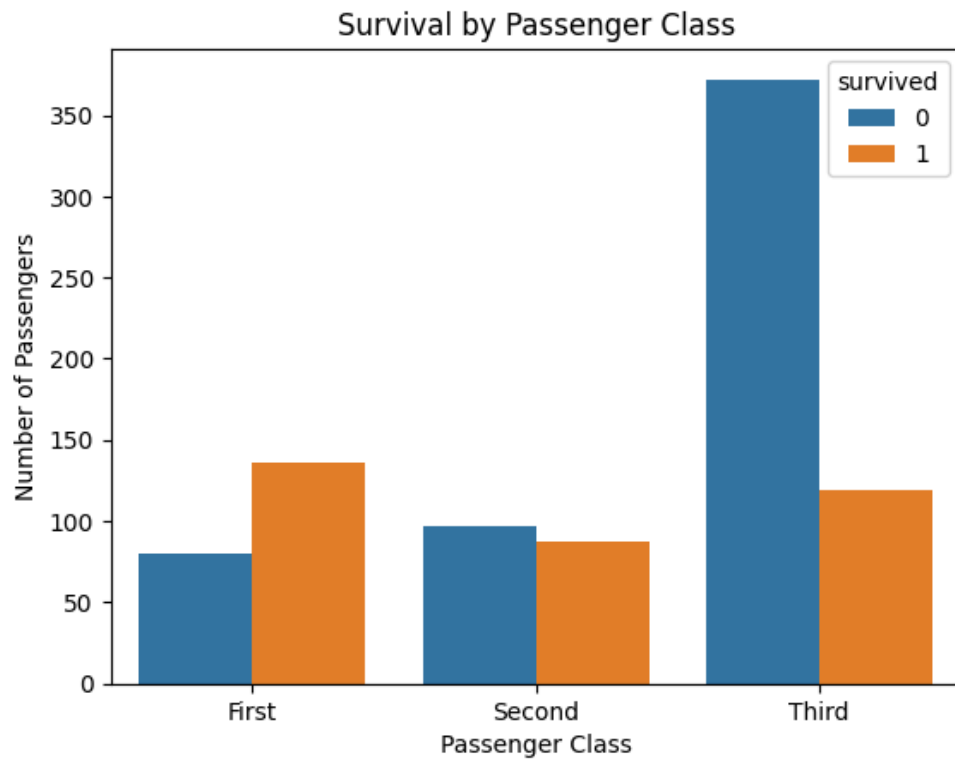
```
sns.countplot(x='class', hue='survived', data=df)
```

```
plt.title('Survival by Passenger Class')
```

```
plt.xlabel('Passenger Class')
```

```
plt.ylabel('Number of Passengers')
```

```
plt.show()
```



```
sns.countplot(x='sex', hue='survived', data=df)
```

```
plt.title('Survival by Gender')
```

```
plt.xlabel('Gender')
```

```
plt.ylabel('Number of Passengers')
```

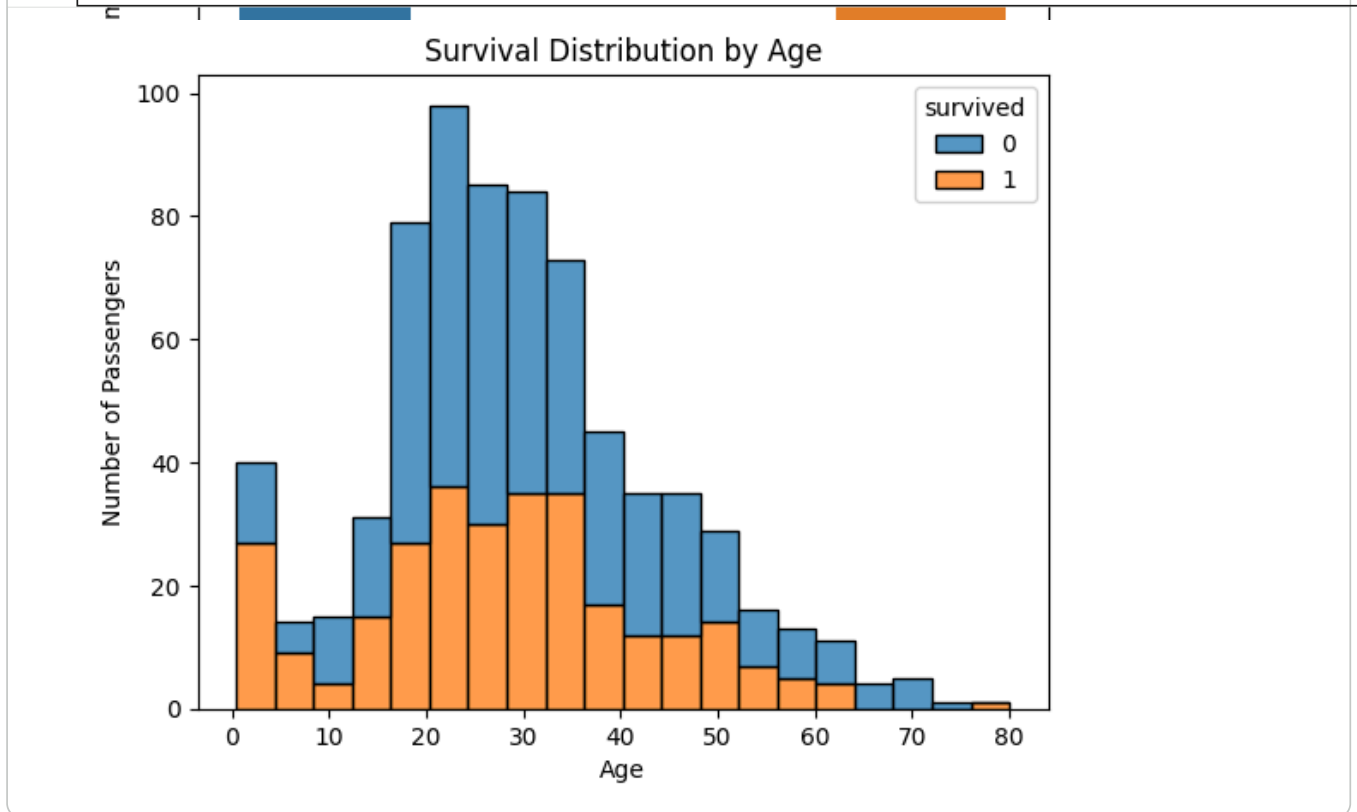
```
plt.show()
```



```
sns.histplot(data=df, x='age', hue='survived', bins=20, multiple='stack')

plt.title('Survival Distribution by Age')
plt.xlabel('Age')
plt.ylabel('Number of Passengers')

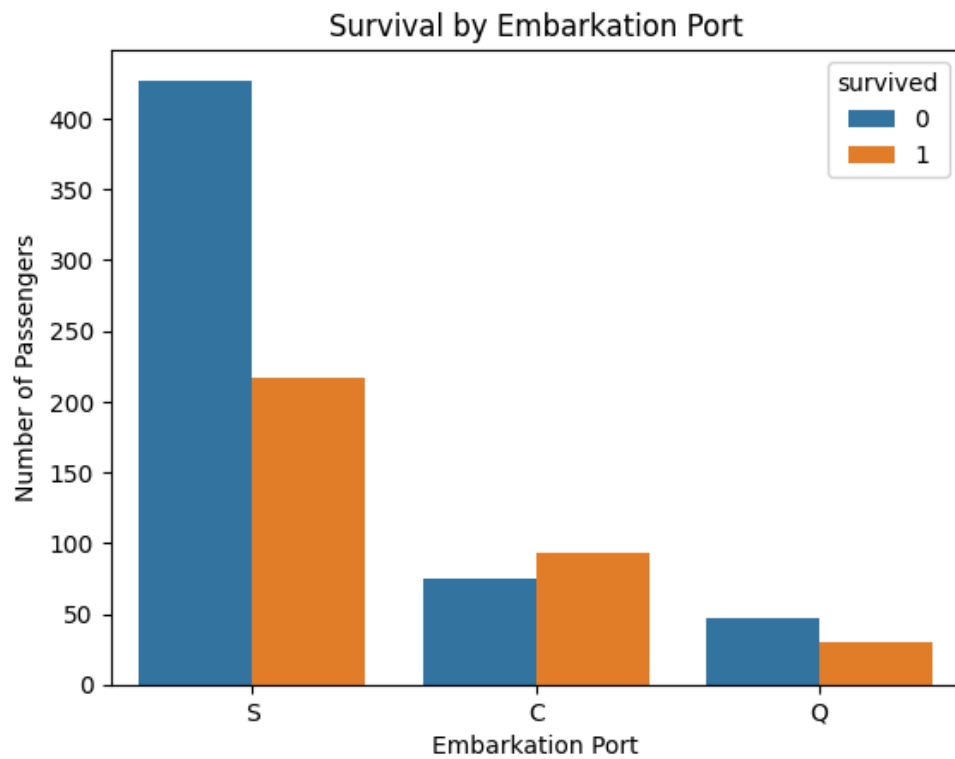
plt.show()
```



```
sns.countplot(x='embarked', hue='survived', data=df)

plt.title('Survival by Embarkation Port')
plt.xlabel('Embarkation Port')
plt.ylabel('Number of Passengers')

plt.show()
```



```
plt.figure(figsize=(8,6))

sns.heatmap(df.select_dtypes(include='number').corr(),
            annot=True,
            cmap='coolwarm')

plt.title('Correlation Heatmap')
plt.show()
```